AdventureWorks 2022 Sales & Customer Performance Dashboard

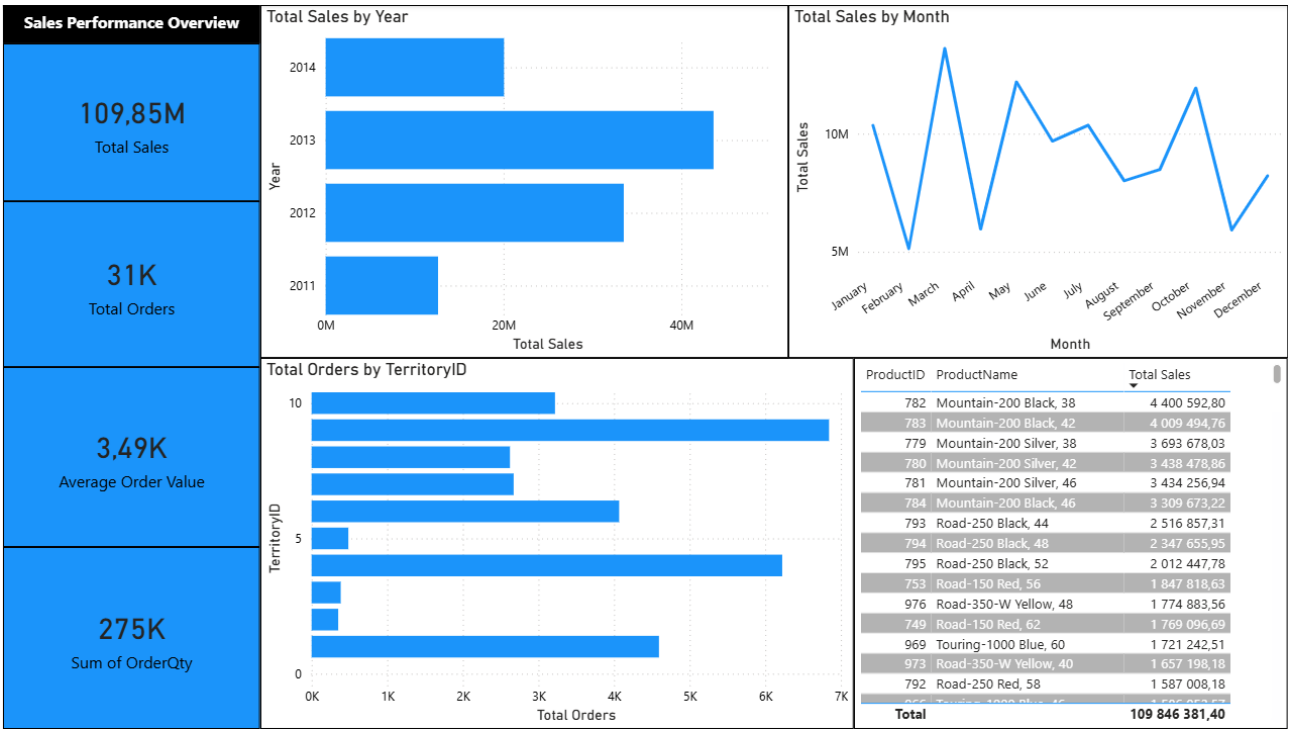
Welcome to the AdventureWorks 2022 Sales Analytics.
This Power BI project provides a deep dive into:

1. Sales Performance Overview
2. Product Performance Analysis
3. Customer Insights
4. Geographic Sales Analysis

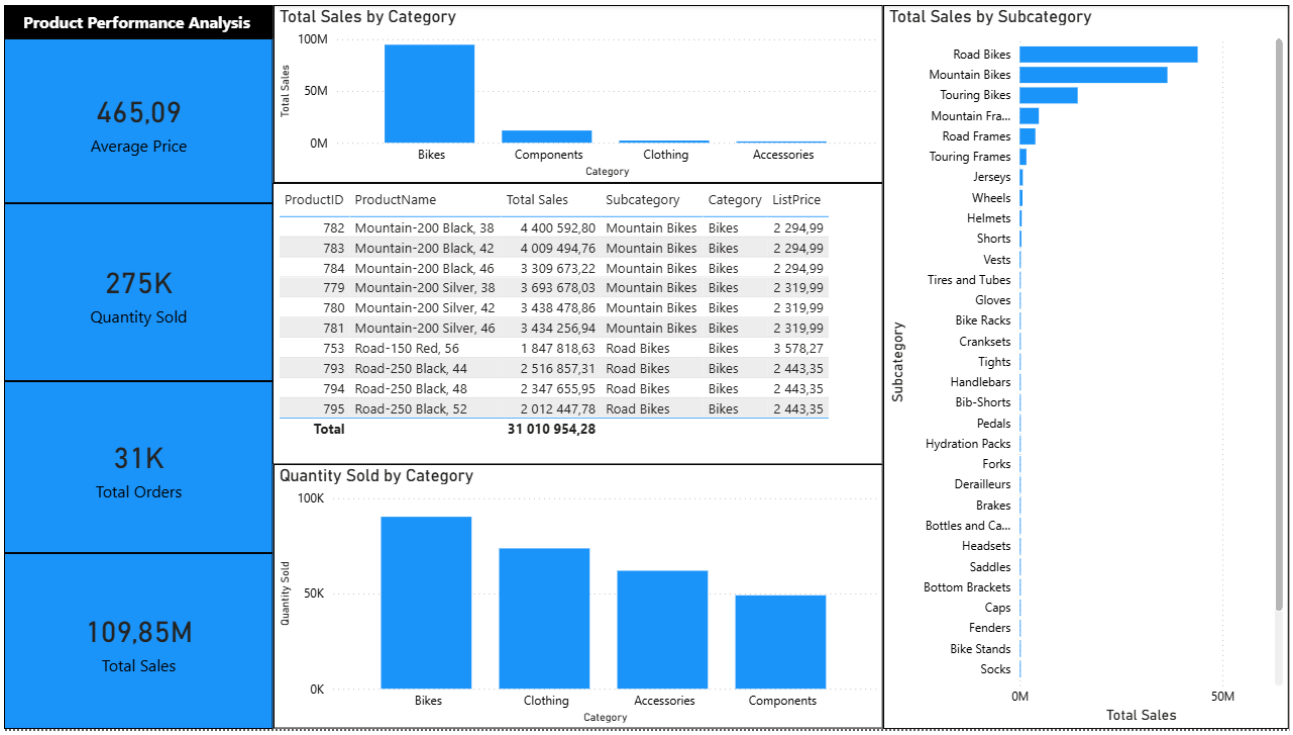
Project Overview

This interactive dashboards analyzes a dataset of product sales between the years 2011 and 2014. It is designed to support business users, analysts, and managers with a comprehensive view of key performance indicators.

1. Sales Performance Overview

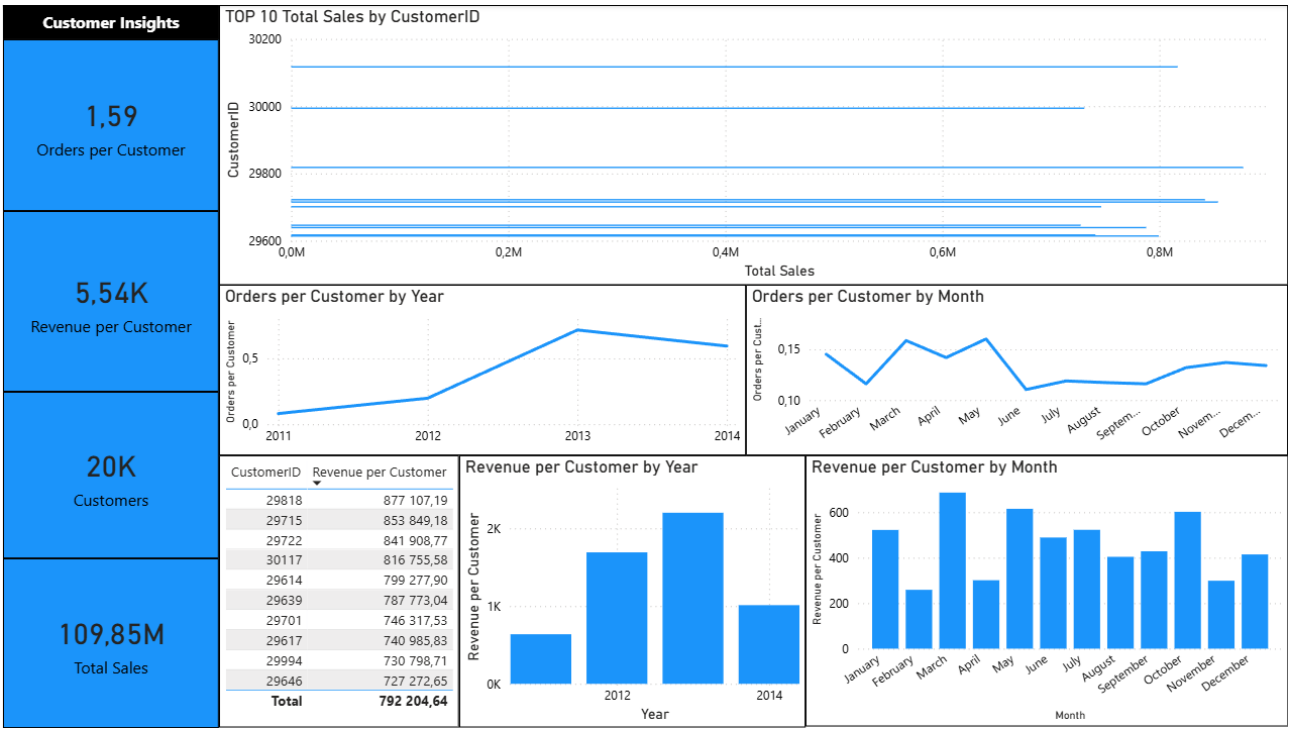


2. Product Performance Analysis



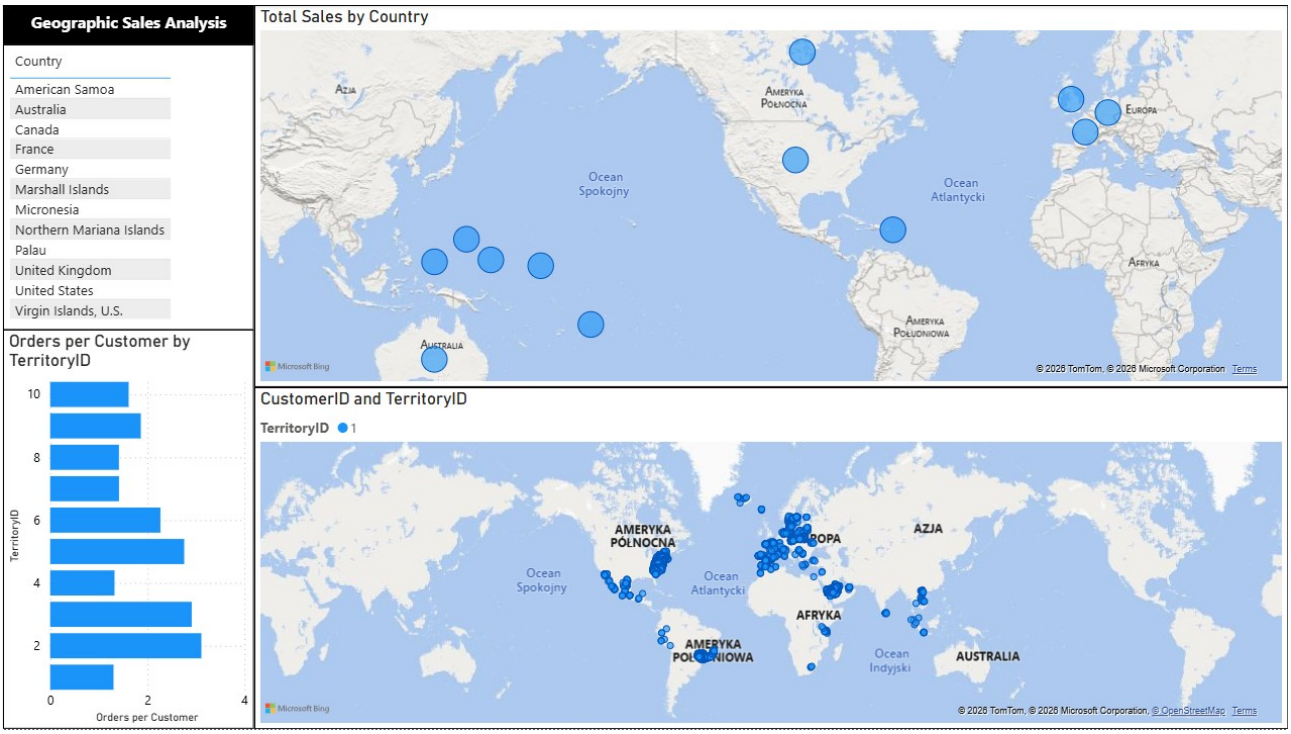
ProductID	ProductName	Total Sales	Subcategory	Category	ListPrice
782	Mountain-200 Black, 38	4 400 592,80	Mountain Bikes	Bikes	2 294,99
783	Mountain-200 Black, 42	4 009 494,76	Mountain Bikes	Bikes	2 294,99
784	Mountain-200 Black, 46	3 309 673,22	Mountain Bikes	Bikes	2 294,99
779	Mountain-200 Silver, 38	3 693 678,03	Mountain Bikes	Bikes	2 319,99
780	Mountain-200 Silver, 42	3 438 478,86	Mountain Bikes	Bikes	2 319,99
781	Mountain-200 Silver, 46	3 434 256,94	Mountain Bikes	Bikes	2 319,99
753	Road-150 Red, 56	1 847 818,63	Road Bikes	Bikes	3 578,27
793	Road-250 Black, 44	2 516 857,31	Road Bikes	Bikes	2 443,35
794	Road-250 Black, 48	2 347 655,95	Road Bikes	Bikes	2 443,35
795	Road-250 Black, 52	2 012 447,78	Road Bikes	Bikes	2 443,35
Total		**31 010 954,28**			

3. Customer Insights



CustomerID	Revenue per Customer
29818	877 107,19
29715	853 849,18
29722	841 908,77
30117	816 755,58
29614	799 277,90
29639	787 773,04
29701	746 317,53
29617	740 985,83
29994	730 798,71
29646	727 272,65
Total	**792 204,64**

4. Geographic Sales Analysis



Key Questions Addressed:

- What is the overall sales performance?
- How have sales changed over the years?
- Which months generate the highest and lowest sales?
- Which territories generate the most orders?
- Which products contribute the most to revenue?
- Which product categories and subcategories perform best?
- What is the average selling price and order value?
- Which categories sell the highest quantity of products?
- Who are the top revenue-generating customers?
- How frequently do customers place orders?
- How does customer revenue change across years and months?
- Which countries and territories generate the most sales?

- Are there seasonal patterns in customer orders and revenue?
- Where are customers geographically concentrated?

Key Metrics:

Total Sales 109.85M

Total Orders 31K

Average Order Value 3.49K

Total Quantity Sold 275K

Average Product Price 465.09

Total Sales 109.85M Overall revenue generated across all customers

Total Customers 20K Size of the customer base

Orders per Customer 1.59 Average purchase frequency

Revenue per Customer 5.54K Average customer value

Highlights & Insights:

Sales Trend by Year

2013 recorded the highest sales (~44M).
2012 performed strongly (~33M).
2014 shows lower sales (~20M) indicating a potential decline or partial year data.
Sales Trend by Month

March recorded the highest monthly sales.
May and October also showed strong performance.
February and November had the lowest sales.
Orders by Territory

Territory 10 and 7 generated the highest number of orders.

Some territories show very low order activity, indicating possible market gaps.

Strong Revenue from Key Customers

A small group of customers generates very high revenue, indicating the importance of customer retention strategies,

Growing Customer Value Until 2013

Customer value and order frequency increased significantly between 2011–2013, showing a strong growth phase.

Seasonal Purchase Patterns

Customer activity spikes in spring and early summer, with lower activity mid-year.

Global Customer Base

Sales come from multiple continents, but Europe and North America dominate.

Tools and Technologies Used:

- **Power BI Desktop** – Dashboard development
- **DAX** – Calculated fields for KPIs
- **SQL Server –** Source data formatting and transformation

SQL QUERY:

Sales Dataset:

```
SELECT
   soh.SalesOrderID,
   soh.OrderDate,
   soh.CustomerID,
   soh.TerritoryID,
   sod.ProductID,
   sod.OrderQty,
   sod.UnitPrice,
   sod.LineTotal,
   p.Name AS ProductName
FROM Sales.SalesOrderHeader soh
JOIN Sales.SalesOrderDetail sod
   ON soh.SalesOrderID = sod.SalesOrderID
JOIN Production.Product p
   ON sod.ProductID = p.ProductID;
```

Product Dataset:

```
SELECT
   p.ProductID,
   p.Name AS ProductName,
   p.ListPrice,
   ps.Name AS Subcategory,
   pc.Name AS Category
FROM Production.Product p
LEFT JOIN Production.ProductSubcategory ps
   ON p.ProductSubcategoryID = ps.ProductSubcategoryID
LEFT JOIN Production.ProductCategory pc
   ON ps.ProductCategoryID = pc.ProductCategoryID;
```

Customer Dataset:

```
SELECT
c.CustomerID,
```

p.FirstName,
p.LastName,
c.TerritoryID
FROM Sales.Customer c
LEFT JOIN Person.Person p
ON c.PersonID = p.BusinessEntityID;

Geography Dataset:

SELECT
sp.StateProvinceID,
sp.Name AS StateProvince,
cr.Name AS Country
FROM Person.StateProvince sp
JOIN Person.CountryRegion cr
ON sp.CountryRegionCode = cr.CountryRegionCode;

Measures (DAX):

```
Total Sales = SUM(Query1[LineTotal])

Total Orders = DISTINCTCOUNT(Query1[SalesOrderID])

Average Order Value =
DIVIDE([Total Sales], [Total Orders])

Month = FORMAT(query1[OrderDate], "YYYY-MM")


Sales Growth MoM =
VAR CurrentSales = [Total Sales]
VAR PreviousSales =
CALCULATE(
  [Total Sales],
  DATEADD(Query1[OrderDate], -1, MONTH)
)
RETURN
DIVIDE(CurrentSales - PreviousSales, PreviousSales)

Quantity Sold = SUM(Query1[OrderQty])

Average Price =
AVERAGE(Query1[UnitPrice])

Customers =
DISTINCTCOUNT(Query3[CustomerID])

Revenue per Customer =
DIVIDE([Total Sales], [Customers])

Orders per Customer =
DIVIDE([Total Orders], [Customers])
```